KIROVAULT

Frontend Development Guide — React + TypeScript

Implementation guide for modular screens, typed UI state, reusable components and a polished responsive desktop interface.

DOCUMENT PURPOSE

This standalone companion document complements the KiroVault Desktop App Blueprint. It follows the Windows-first, local-first product direction. Recommendations are proposed guidance, not claims that these features already exist.

Contents and implementation guidance

Module structure

• src/app: App.tsx, router, providers and error boundaries.

• src/components/shell: AppSidebar, TopBar, breadcrumbs, command bar, PreviewInspector.

• src/components/files: FileCard, FolderCard, FileGrid, FileList, context menu and selection toolbar.

• src/components/preview: PreviewRouter and renderer per file family.

• src/features: home, explorer, gallery, timeline, search, storage, favorites and settings.

• src/services: typed wrappers for filesystem, indexing, thumbnails, metadata and search.

Component contracts

• FileCard receives a typed file record, selection state and action callbacks; it does not perform filesystem work itself.

• PreviewRouter chooses a renderer from detected kind/extension and always provides a fallback.

• Inspector renders preview status, metadata and actions; single selection must not launch an external app.

• Shared empty/loading/error components keep behavior consistent across screens.

State management

• Separate navigation state, selection state, view preferences, preview state and scan state.

• Persist safe user preferences; do not store file contents in global state unnecessarily.

• Cancel stale thumbnail requests when folder or selection changes.

• Use virtualization for long lists and lazy-load only visible/near-visible thumbnails.

Design implementation

• Create tokens for surfaces, spacing, radius, typography, elevation, category colors and motion.

• Use a three-zone layout with collapsible sidebar and hideable inspector.

• Support light/dark/system theme, high contrast and reduced motion.

• Ensure focus indicators, keyboard operation, accessible names and clear disabled states.

Frontend acceptance tests

• Navigation back/forward/up and breadcrumb updates correctly.

• Grid/list switching preserves folder and sensible selection behavior.

• Keyboard multi-select and context actions match the current selection.

• Unsupported preview shows a useful reason and Open with action.

• Thousands of entries do not render all DOM rows or freeze the UI.

References

• React documentation: https://react.dev/learn

• TypeScript Handbook: https://www.typescriptlang.org/docs/handbook/intro.html

• Vite guide: https://vite.dev/guide/

• WAI-ARIA Authoring Practices: https://www.w3.org/WAI/ARIA/apg/

• KiroVault Desktop App Blueprint, pp. 16–17 (user-provided source).

Reference note: external documentation may change. Verify current versions, licensing, APIs and platform prerequisites during implementation.